PART II — FROM PHYSICAL TO DIGITAL · CHAPTER 07

Interfaces as Tolerances

07

The limits that shape interaction

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When you work in industrial design, you soon discover that a user's interaction with a product begins before the object exists. It begins with knowing which materials you can source, which machine you will produce on, and which mold you can afford. Many of the sensations we later credit to design are decided there: how firm or soft a grip feels, where one part ends and another begins, what it costs to repeat the object ten thousand times.

A toothbrush shows that whole journey, because it is an everyday contact between hand, material and body. Almost nobody thinks about its mold, and yet every manufacturing decision reaches the mouth of the person using it. I use it as an example of something anyone can check against the object in their own bathroom.

The question I want to carry from there into the digital world is simple. First: what can we make? Then: how can someone use it? And finally, the one that weighs most once the product is in a person's hands: what are we demanding of them when they use it every day? Chapter 5 introduced digital materials and Chapter 6 their preservation over the life of a product; this one examines how those decisions reach the person through the interface.

What a toothbrush allows

[4/5 format — Unbranded toothbrush with a rigid body, soft grip zones and bristles on a neutral surface]
Figure 7.1 — Rigid body, soft grip zone and bristles: three decisions the user feels as one.

Consider a toothbrush made of a rigid body and softer grip zones. That combination is not decided on the drawing alone. Multi-component injection, the mold and the compatibility between materials shape its geometry, its joints and its surface: where the soft part can end, what thickness it allows, how it bonds to the body and what texture remains visible. The machine bounds what is possible; the design team still makes decisions inside those bounds.

It helps to separate that body from what happens with the filaments. The documentation from ZAHORANSKY, a manufacturer of brush-making machinery, distinguishes the injection of one- or multi-component parts from the placement of the filaments and their finishing, which includes trimming and rounding the tips.[1] They are different processes and each one delivers different decisions: the first defines the grip and the shape; the others define what touches gum and enamel. Knowing which one a decision belongs to keeps us from blaming the mold for what is the filament's, and the filament for what is the mold's.

The same goes for hard, medium and soft bristles. The commercial categories are useful for talking about perception, but the real response depends on a combination of material, diameter, length and arrangement of the filaments. Two brushes that both say “medium” can feel different. I am not comparing specific brushes here; I use the categories only to show that a sensation is the consequence of several variables at once.
To follow the argument it helps to fix three words with a simple meaning. They resemble one another, which is why they get confused.

TABLE 7.1 — THREE TERMS THAT ARE NOT SYNONYMS

TERM	WHAT IT MEANS IN THIS CHAPTER	IN THE TOOTHBRUSH
Constraint	A condition that bounds a decision, such as the available process, the space of a room or the budget	The mold you can afford and the materials that are compatible with each other
Technical tolerance	Permitted variation from a specification, for example in a dimension or in the fit between parts	The admissible variation in a wall’s thickness or in the fit between the body and the soft zone
Acceptance threshold	The limit of an admissible response, such as actuation effort, the intensity of an alert or the duration of a task	How much firmness the user tolerates before it becomes bothersome

This chapter borrows the idea of tolerance from manufacturing and extends it to interaction design. The availability of a material is a constraint; its properties and permitted variations help establish its operating margin. People also show differences and limits that must be studied. That extension is an editorial proposal of this chapter: they are not technical synonyms and I do not use them as if they were.
Let us push the logic to the extreme. With the right machinery and materials it is possible to manufacture a toothbrush with metal bristles. Nothing in the process prevents it. And yet being able to make it does not show that the combination is appropriate for the intended contact: that a way to produce something exists does not answer whether it should exist. This is the point at which the possibility of production stops being enough to justify a design.
The user feels the consequence of these decisions every time they pick up the object and use it. In an application something similar happens, even if its materials are less visible.
Being able to make a toothbrush with metal bristles does not show that the combination is appropriate for the intended contact.

One route, three interfaces

[3/2 format — Car dashboard with a screen showing a schematic route]
Figure 7.2 — Car: attention is shared with driving.
[3/2 format — A hand holds a phone showing a schematic route while walking down a street]
Figure 7.3 — Phone: one hand occupied and daylight on the screen.
[3/2 format — Tablet on a table showing a schematic route, next to a notebook]
Figure 7.4 — Tablet: unhurried browsing to plan a trip.
Take a need everyone recognizes: finding your bearings and getting somewhere. A navigation service such as Google Maps serves it in a car, on a phone and on a tablet, and the need is the same in all three cases. What changes is everything surrounding the person who has it. I do not promise that every current version of the service offers exactly the same functions on every device; I use the example to compare scenarios, not to describe a product.

In a car, attention is shared with driving. Google's design guidance for driving establishes that driving-related tasks take priority over any other, that information must be readable at a glance — it proposes a maximum of two seconds to read a piece of content and return to the road — and that no operation should require taking both hands off the wheel.[2] These are limits of context, not of screen: the same route calls for fewer steps, shorter instructions and voice.

On a phone, during a walk, the hand is occupied, daylight competes with the screen, battery matters and attention is divided with the surroundings: the street, other people, the next crossing. What the interface must achieve is that the next action is recognizable and can be handled with one hand.

The tablet, by contrast, tends to appear when someone is preparing a trip. There is more surface and room for unhurried browsing: comparing alternatives, seeing more context. But it is wise not to assume that every tablet has the same sensors or serves the same purpose as another.

TABLE 7.2 — THE SAME NEED IN THREE SITUATIONS

SCENARIO	INTERACTION CONDITIONS	DECISIONS IT RAISES
Car in motion	Attention shared with driving, reach of controls and brief reading	Prioritize directions, reduce steps and make use of voice and audio
Phone during a walk	Occupied hand, outdoor light, battery and attention to the surroundings	Make browsing and one-handed use easy; make the next action recognizable
Tablet when planning a trip	Larger surface and room for unhurried browsing	Compare alternatives and show more context without assuming every tablet has the same sensors

That is why reducing the difference between these three scenarios to screen inches is a simplification. The device changes the available channels and also the effort that can be asked of the person; at bottom, it changes their acceptance threshold. And the interface is transformed even more when the interaction occupies the whole body.

The body occupies the interface

[transparent object — Exploded view of a conceptual multisensory seat: seat and backrest, base and actuators, supports, vibration and sound, optional thermal system, and covers]

Figure 7.5 — Conceptual multisensory seat for a 4D theater, exploded view. The six points correspond to Table 7.3.

To see this same problem in a physical object, I propose a conceptual multisensory seat for a 4D theater. It is a working example, not a portrait of a commercial installation: it lets us analyze movement, vibration, thermal effects and coupling with sound without attributing this whole configuration to every theater that exists, or

assuming that each effect originates inside the seat. Part of the sound, the air or the heat may come from the room.

Taken apart, the seat shows that each part carries a design decision and that each decision has an aspect someone must evaluate.

TABLE 7.3 — A SEAT BROKEN DOWN INTO DECISIONS

PART OR SYSTEM	DESIGN DECISION	ASPECT TO EVALUATE
Seat and backrest	Geometry, support and padding	Fit to different bodies and comfort throughout the show
Base and actuators	Travel, speed and acceleration	Controlled movement and clear space required
Contacts and supports	Position of arms, feet and hands	Stability, access and the possibility of changing posture
Vibration and sound	Intensity, duration and synchronization	Clarity of the effect and accumulated sensory load
Thermal system, if any	Location, control and exposure time	Contact, regulation and safe conditions of use
Covers and joints	Materials, cleaning and access to components	How the experience holds up during operation

All these variables need specification, testing and acceptance criteria. An intention of immersion does not by itself guarantee an experience that suits everyone: the backrest that supports one body well can leave another without support, a travel one person enjoys can bother another, and a vibration that is clear at the start of the show can turn into accumulated sensory load by the end. Here the idea of tolerance returns. The point is not that the seat works on average, but knowing within which variations it still works and for whom.

But the seat still has to be installed in a room, and there limits appear that the design of the seat alone cannot resolve.

The room designs too

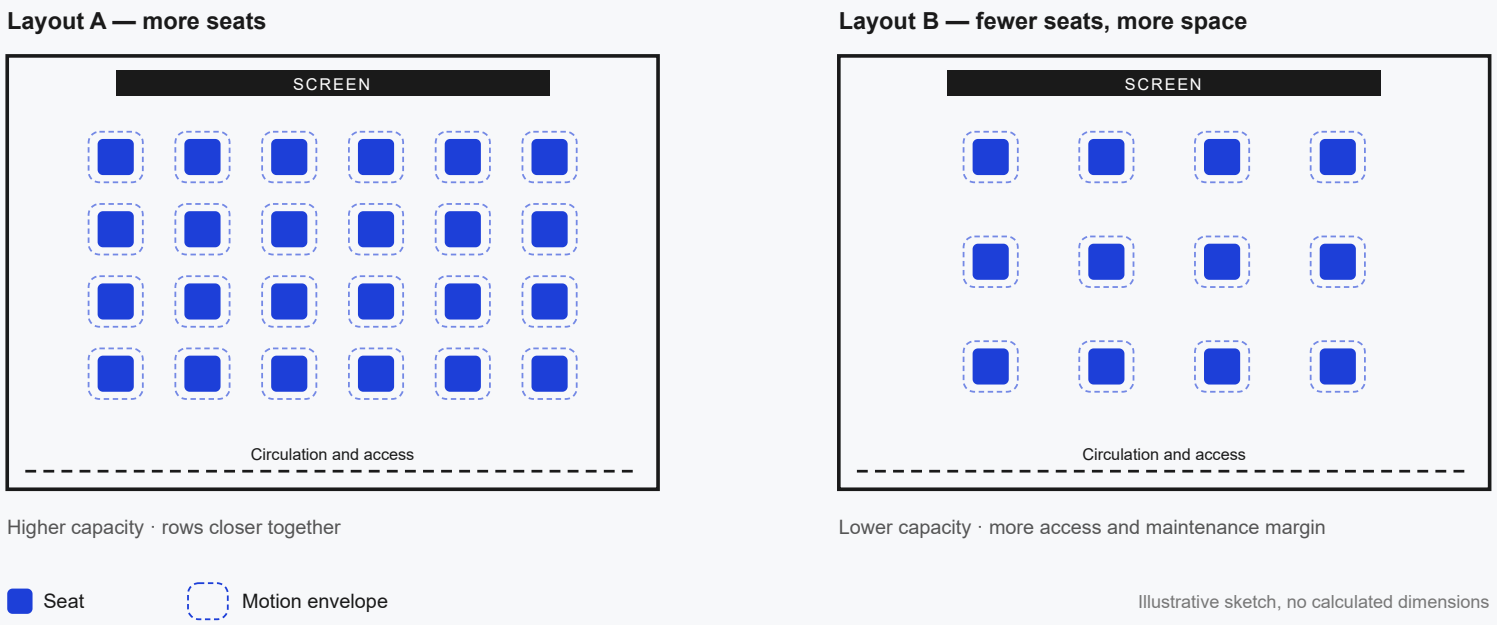


Figure 7.6 — Two hypothetical configurations of the same room. Illustrative sketch: it does not represent calculated dimensions or a specification.

Suppose the commercial goal is to have a certain number of seats. That number translates into a conversation with the available floor area, the spacing between rows, the motion envelope of each seat, the

access routes and the space for maintenance. Accessibility and evacuation have to be considered from the outset and not at the end; the applicable dimensions depend on the project and its location, so I do not propose any here.

A seat that works well as a prototype can take up too much space, or cost too much, once many units are installed. The consequences chain together. Changing the width alters the capacity. Changing the base alters the movement and the maintenance. Reducing the space between rows changes access and the experience of the person seated. Each adjustment that resolves one limit pushes the problem toward another.

It helps to distinguish two volumes that are easily confused. Units manufactured affect the investment in tooling and the cost per part. Seats installed affect the capacity and revenue of the service. Whoever produces the seat thinks about the first; whoever operates the room, about the second; and the design has to speak to both.

Nor is the profitability of a room explained by counting seats. It depends on occupancy, price and operation. More seats do not mean more revenue if the experience forces occupancy down or makes maintenance more expensive.

In digital products, costs are also spread over an expected capacity of use. That capacity must be analyzed per platform and per function.

Every platform has its own math

Back to the screens: mobile, tablet, television, smartwatch. A function can be technically feasible and still not add enough incremental value to sustain its development, its testing and its maintenance on a given platform. These are two different questions and it pays to ask them separately: what does the hardware allow, and what does the economics justify.

The watch serves for the first: brief interaction, small controls, limited battery and capabilities that vary between models. On television, reading distance, the remote control and the diversity of devices all count. Those are constraints of hardware and context. That a function is not built there for reasons of cost is another kind of constraint, and mixing them leads us to justify with technology what is really an economic decision, or the reverse.

The audience that matters is that of this product and this function on that device, not that of the platform in general. It is not correct to say that watches or televisions always have insufficient critical mass: a small audience can justify a high-value solution, and a large audience may not use the function.

600,000	120	5,000
EQUIVALENT ANNUAL COST IN MXN, ANNUALIZED DEVELOPMENT AND SUPPORT · HYPOTHETICAL	MXN NET INCREMENTAL CONTRIBUTION PER YEAR PER USER OF THE FUNCTION · HYPOTHETICAL	FUNCTION USERS NEEDED TO COVER THE COST · HYPOTHETICAL

With those assumptions the calculation is a division: 600,000 divided by 120 is 5,000. What matters is what gets counted. It is the users who use the function, not downloads or total users, which neither substitute for effective adoption nor demonstrate that contribution.

When the value of the function is an operating saving, retention or a service obligation, the case is built with that evidence and not with a per-user calculation. And there is a limit no calculation moves: the necessary safety and accessibility are not removed by a popularity calculation.

Beyond being able to build and sustain the function, we need to understand how much change the person receiving it will accept.

PART II — FROM PHYSICAL TO DIGITAL · CHAPTER 07 · 06 ACCEPTANCE

Advancing as far as people accept

That question has a name in industrial design: MAYA, for *Most Advanced Yet Acceptable*, a phrase associated with Raymond Loewy. I use it as an orientation for balancing novelty and familiarity, not as a formula. The study by Hekkert, Snelders and van Wieringen examines how typicality and novelty jointly contribute to aesthetic preference for industrial products.[3] Its scope is that: aesthetic preference. It does not let us conclude that MAYA, by itself, predicts adoption, health or economic return.

What I propose is to apply it to functions. Keep the main benefit of a function and adapt its complexity to the device, the task and what the user can recognize and learn. A watch can offer a brief action and transfer the extended configuration to the phone, if the continuity between the two is understandable and useful. If it is not, the watch has only added a step.

The decision to include, simplify, transfer or discard a function combines four questions. Is it safe and accessible for the intended use? Can the device sustain it? Does the user understand and accept the interaction? Does the value justify sustaining it? MAYA contributes to the third, and only to it. That combination is the method this chapter proposes; it is not an extension attributed to Loewy.

That acceptance also depends on small decisions that repeat, like a connection sound.

PART II — FROM PHYSICAL TO DIGITAL · CHAPTER 07 · 07 SOUND AND REPETITION

Sound takes up space too

A concrete example of a branded sound signal is found in the Marshall Major IV headphones. Its support page, describing how to pair them over Bluetooth, states that the brand's pairing sound should be heard.[4] From that fact, and only from it, we can analyze the function of a recognizable confirmation: it communicates

a state (“it’s connected”) and builds identity. I do not extend that behavior to any other model, to a wired connection or to other events without verifying it.

The difference matters between an occasional alert and repetition across a working day. Hearing a signal once, when unboxing a device, is one thing; hearing it dozens of times a day is another. Intensity, duration, frequency, context and user control are all part of the design decision. Whether a sound pleases or annoys is not a universal characteristic: it is a response that needs research, with different people in different situations.

[16/9 format — Unbranded over-ear headphones next to a phone, with a small blue indicator light on]

Figure 7.7 — Confirmation: a useful signal communicates a state.

[3/2 format — An unbranded smartwatch photographed with long exposure, its screen lighting up several times over a single night]

Figure 7.8 — Repetition: the same signal, many times in one day.

[3/2 format — Three trays of sorted blank slips of paper, one of them with a blue marker]

Figure 7.9 — Grouping: not every alert weighs the same.

[1/1 format — Close-up of a volume knob and a switch on an unbranded device, with a blue mark]

Figure 7.10 — Control: the person decides what sounds, when and how loudly.

The same holds for an application’s notifications, sounds and vibrations. Choosing categories, schedules, frequency and channels, grouping messages and distinguishing among confirmations, promotions and critical alerts are decisions that distribute effort: either the product takes it on, by sorting and filtering, or the person does, by muting and dismissing. Personalization is, at bottom, the control the user has over their own experience.

That control includes the exits. Deleting data and closing an account are part of the same relationship and call for clear consequences and an understandable interaction: whoever decides to leave should know what is kept, what is erased and when. It is also worth distinguishing errors, task recovery and frustration, because they are not the same: an error corrected in one step does not weigh like one that forces you to start over.

And there is a trap in the metrics. Studying who leaves, who mutes and who keeps using the product despite the annoyance tells us more than counting uses. Frequent use, by itself, does not demonstrate acceptance: some people keep using what bothers them because they have no alternative.

In a work product, the person may have even less capacity to decide whether to keep using it.

Scanning all day long

[4/5 format — Three unbranded code scanners with different grips on a work table next to a pair of gloves]

Figure 7.11 — Alternative grips and capture methods: comparison starts from the task, not the catalog.

Let me pose a case. A manufacturing logistics operation equips its staff with phone-like capture terminals to read QR and bar codes: at raw-material receiving, at finished-goods dispatch and in logistics tracking. The convergence of screen, connectivity and capture in a single device can bring real value. For a specific project it will be necessary to verify why that device was selected and not another; I take no reason for granted here.

Let us compare that device with a traditional scanning gun, but starting from the task and not the catalog: the grip, the weight distribution, the activation, the orientation of the reader, the confirmation that the capture happened, use with gloves and the need to look at the screen. The question I want to leave open is this: by concentrating functions in the device, did we also change the effort we ask of the user?

I am not starting from zero. OSHA, the United States occupational safety and health agency, identifies among the ergonomic risk factors of warehouse work awkward postures and the repetition of the same tasks, and notes that exposure to those factors increases the risk of musculoskeletal disorders.[5] That evidence justifies evaluating exposure. It does not show that every phone-shaped terminal causes injuries or that any gun is superior; the guidance does not compare those devices.

How, then, do we evaluate it? With a comparison in representative conditions, with the participation of workers and of ergonomics specialists. We observe the capture and correction tasks, the duration of the grip, postures, changes of hand, repetition, errors and the discomfort people report. And we include alternatives according to the operation: grips, mounts or readers that leave the hands free. One condition: early tests must not require anyone to endure discomfort in order to obtain data.

Software intervenes too. Additional confirmations, read errors, lost connections or unnecessary screens can lengthen the task and change how the device is held. It pays to separate the time saved in one capture from the effect of repeating the cycle across the shift: what looks like a detail in a short test can be what weighs most at the end of a full working day.

A warning remains. Those who implement technology in an operation tend to think about possible claims only when they arrive. I do not predict lawsuits, nor do I claim that a device that has not been studied causes harm. What I hold is a demand for responsible anticipation: **we do not need to wait for a claim to investigate an avoidable exposure.**

Measure before asking people to adapt

This journey can be turned into a method. I propose a map of limits for each relevant interaction. The map records the task and its context, the variable being observed, the acceptance criterion, the available evidence, who validates that criterion and the action to be taken if it is not met. When no supported limit exists, none is invented: the uncertainty is recorded, along with how it will be investigated.

TABLE 7.4 — SIX LAYERS OF THE MAP OF LIMITS

LAYER	EXAMPLE VARIABLE	EVIDENCE NEEDED TO DECIDE
Material and process	Variation in a part or grip response	Specification and manufacturing tests
Device and context	Actuation, reading and steps during the task	Tests on the intended device and in the intended situation
Sensory and cognitive	Alert frequency, errors and recovery	Research with diverse users and representative exposure
Space and operation	Spacing, movement and access	Validation of the installed whole
Economics	Effective use, cost and incremental value	Scenarios and adoption tracking
Health	Postures, force, repetition and duration	Competent ergonomic evaluation and response to reported discomfort

Identify, measure, decide, validate

The method repeats every time the hardware, the flow or the pace of work changes.

01
Identify
Recognize the exposure.

02
Measure
Measure the response.

03
Decide
Decide the adaptation.

04
Validate
Validate sustained use.

Identify the exposure. Before measuring we need to know what the person is exposed to: which gestures they repeat, for how long, in what posture and at what pace. This is the step most often skipped, because it seems obvious in the demo and stops being obvious during the shift.
Measure the response. Measure in representative conditions and with the people who will use it, with diversity of bodies and experience. A favorable average can hide relevant differences between people, which is why it pays to look at the spread and not only the center.
Decide the adaptation. The decision rests on verifiable requirements and evidence. It may mean changing the object, the flow or the operation; what it cannot be is simply passing the adaptation on to the person using it. A commercial improvement does not automatically offset an unmet health criterion either.
Validate sustained use. Real use, week after week, is the test a demo cannot give. The conditions for modifying or stopping an implementation are defined, and it is evaluated again after changes in hardware, flow or pace of work.

What the user ends up paying

That person
— Closing of the chapter

Back to the toothbrush. The metal bristles made an inadequate decision visible: nobody needs a study to see that they should not touch a gum. In a digital or hybrid product the effort is not so easy to see, because it is spread across alerts, gestures, grips, waits and corrections, each small and all repeated.

The responsibility belongs to the product. It means bringing health and everyday exposure into device selection, into the definition of requirements and into the acceptance of the implementation. A successful demo needs to be complemented with evidence of representative use. And when tests reveal an avoidable load, what is called for is to revisit the object, the flow or the operation before passing on to the person the obligation to adapt.

As product managers, we have to look at what happens after the first interaction. That device will stay in someone's hand throughout their working day, with the same gestures and the same demands we defined when we implemented it. Chapter 8 follows that trail: it is devoted to what changes when a product scales.

REFERENCES

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5. **Occupational Safety and Health Administration (OSHA)** — Warehousing, Hazards and Solutions. Official guidance on ergonomic risk factors in warehouses; it supports the need to evaluate and does not compare the devices discussed. ↩
6. **This book** — Chapters 5 (digital materials: APIs, data and constraints) and 6 (designing for maintenance, not only for launch).
7. **Editorial note** — the proposed structure, the scenarios, the conceptual seat, the room and the economic example are the author's elaboration and do not represent results from real projects. The

figures in the economic example are hypothetical.

← CHAPTER 06

Designing for Maintenance, Not Just Launch

CHAPTER 08 — COMING SOON →

Part II continues

J. Angel Sosa — Thinking in Products, Building Systems

All chapters · RSS · Privacy

Thinking in Products, Building Systems

Part I — Foundations

1. 01What Can Industrial Design Teach?
2. 02Product Strategy: Materials & Cost of a Decision
3. 03What a Production Line Teaches About Architecture
4. 04The Product Is Never Just the Object

Part II — From Physical to Digital

5. 05Digital Materials: APIs, Data and Constraints
6. 06Designing for Maintenance, Not Just Launch
7. 07Interfaces as Tolerances
8. 08Chapter 08 — coming soon